

STUDENT ACTIVITY FILE

INNOVATIVE DESIGN LAB - I

Academic year: 2026-2027

Department: Computer Science and Engineering

Section:

Innovation Ecosystem Project ID: SAIRAM-IDL1-2026-CSE-09

Title of the Project: CYBERFIDEL: Autonomous Hybrid-AI SOC Analyst & Investigative Engine

Inter disciplinary Project : ☒ Yes ☐ No **Team**

Members (Max. 3)

S.No	Name	Student ID
1	Gokula Kannan M	SEC25CS196
2	Lakshan M	SEC25CS036
3	Sivabalan T	SEC25CS101

Project Version Is this project:

☐ Version 1 – New Project ☐ Version 2–Pass out Student Project(Carried Forward)

(If Version 2, provide the following details of the original project):

Previous Project Title	Sairam Innovation Project ID

Domain Name: Cybersecurity & Artificial Intelligence

IEEE Society: IEEE Computer Society

IEEE Community: IEEE Student Branch Sairam

Club and Cells: Sairam Innovation Ecosystem / Cyber Security Club

Name of the department IDL - I Coordinator: Dr. A. SHEELA

Faculty ID: IDL-FAC-01

Name of the Supervisor: Dr. A. SHEELA

Faculty ID: IDL-FAC-02

Name of the External Guide: Cyber Forensic Specialists & Industry Leads

Designation & Organization: Cyberdome Investigations / Zoho Corporation

SDG	Goal Number With Name	SAP Code with Explanation
Primary SDG:	SDG 9: Industry, Innovation, and Infrastructure	1. SAP-SDG9-CYBER-01: Resilient cyber infrastructure & AI SOC automation. 2. SAP-SDG9-CYBER-02: Zero-trust cloud-edge security architecture. 3.
Secondary SDG:	SDG 16: Peace, Justice, and Strong Institutions	1. SAP-SDG16-SEC-01: Evidence privacy & chain-of-custody enforcement. 2. SAP-SDG16-SEC-02: Citizen PII protection in digital investigation. 3.
Tertiary SDG:	SDG 8: Decent Work and Economic Growth	1. SAP-SDG8-AUTO-01: Reduction of SOC analyst alert fatigue. 2. SAP-SDG8-AUTO-02: Decreasing MTTR from 45 mins to < 30 seconds. 3.

S.No.	Title	Start Date	Completion Date	Mark (Out of 10)	Remarks	Signature of the IDL - I Coordinator
1	Orientation & Entrepreneurial Mindset	01/08/2026	07/08/2026		Completed	
2	Problem Identification & Passion CV	08/08/2026	14/08/2026		Completed	
3	Customer Segmentation & Persona Creation & JTBD	15/08/2026	21/08/2026		In Progress	
4	Ideation, Market Size & Competitor Analysis	22/08/2026	28/08/2026		Upcoming	
5	Milestone1:Problem-Solution Fit Presentation	29/08/2026	04/09/2026		Upcoming	
6	MVP Design & Prototyping					
7	Opportunity Assessment & Feasibility					
8	Business Model & Lean Canvas					
9	Literature survey					



10	Milestone 2 : Business Model Presentation					
11	Marketing & Sales Strategy (GTM)					
12	Financial Planning & Team Building					
13	Milestone 3 : Final Pitch- Venture Feasibility					

Outside World Projection

Details of the Competition Attended (Hackathon / Idea Pitching)

S.No	Name	Date	Place	Remark	Signature
1	Hac'KP 2026 (7th Edition National Hackathon by Kerala Police Cyberdome)	14/08/2026	Zoho Corporation, Chennai		Individual Hackathon Finalist (Sivabalan T)
2					

Design Patent / Idea Patent Publication / Conference Paper

1	Confidentiality-Preserving Autonomous SOC Triage via Tokenized Edge-Cloud Multi-Agent Systems	August 2026	IEEE / USENIX Track		Paper Manuscript & Patent Drafted
2					

IDL – I Coordinator
Dr. A. SHEELA

Creative Innovator
Gokula Kannan M

H.O.D
Head of Department



புயன்நாள் புதுபாத்து
எதுவும் இல்லை.

சென்னை
M. Ramesh

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PHOTOS / LITERATURE PAPERS / PATENT SEARCH / SKETCHES (PASTE OR ATTACH SEPARATE PAGES)

WEEK – 1

Security Operations Centers (SOCs) generate over 10,000 security logs daily, creating severe alert fatigue where human analysts spend 30?45 minutes manually triaging single events. Current AI supports cannot be deployed widely due to data privacy regulations and risks of uploading confidential network payloads to external servers.

1. Orientation & Entrepreneurial Mindset

The proposed solution, SENTINEL, introduces Reversible Tokenized Pseudonymization, scrubbing 100% of sensitive PII, IP addresses, and hostnames locally before AI reasoning occurs. This enables enterprise-grade autonomous triage at \$0 operational cost for 90% of routine alerts using local consumer GPU hardware (NVIDIA RTX 3050).

• Understand entrepreneurship & innovation

The global SIEM automation and threat intelligence market is valued at \$13.2 Billion+ with an 18.5% CAGR. Primary target markets include Enterprise SOCs, Managed Security Service Providers (MSSPs), Air-Gapped Defense Networks, Cyber Crime Units, and Startups.

- Technical skillset is complemented by providing 700+ parameters and down to 4-bit (168 MB) CSV format to execute high-capacity intelligence on edge workstations.

- Startup journey from idea to venture.
 1. Problem Formulation: Identifying SOC alert fatigue and data privacy constraints.
 2. Proof-of-Concept: Engineering local Zero-Trust Data Sanitizer (src/sanitizer.py).
 3. MVP Architecture: Building 3-Tier Hybrid AI Router (src/router.py).
 4. Validation & Pitching: Selected for National Pitching Round at Hac'KP 2026 @ Zoho Corporation.
 5. Commercialization: Deploying open-source SaaS framework and enterprise MSSP engine.

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WEEK – 2

A. Personal Interest & Cyber Defense Motivation:

- Deep Interest in System Architecture & Defensive Security. As Team Lead (Gokula Kannan M), my primary focus lies in solving critical engineering bottlenecks where human manual triage speed fails to match machine-speed cyber attacks.

B. Problem Formulation - SOC Alert Fatigue & Telemetry Privacy Dilemma:

- The Alert Triage Bottleneck: Modern Security Operations Centers (SOCs) ingest over 5,000 daily security logs from SIEM agents (Wazuh, Elastic). Tier-1 human analysts spend 30 to 45 minutes investigating single alerts, causing ~70% of security notifications to remain completely unexamined.

- The Cloud Data Leakage Threat: Transmitting un-scrubbed raw telemetry-containing internal IP topographies (RFC 1918), officer emails, hostnames, and credentials-to commercial cloud AI models creates catastrophic data leakage risks, violating privacy mandates such as India's DPDP Act 2023.

C. Project SENTINEL Application & Value Proposition:

- Autonomous Zero-Trust AI Co-Pilot: Delivering machine-speed incident triage (<30s MTTR) via an in-RAM regex data sanitizer, 3-tier model router, and AST code sandbox, ensuring zero cloud telemetry leakage and \$0 Tier-1 compute cost.

A. Technical Skills & Core Competencies (Gokula Kannan M - Team Lead):

- Systems Architecture & Venture Strategy: Expertise in high-throughput asynchronous backend engineering (FastAPI, Python), multi-tier AI model cascading (Ollama local GPU to cloud fallback), and technical project governance.
- Defensive Cybersecurity & Threat Intelligence: Applied knowledge in SIEM rule correlation (Wazuh), MITRE ATT&CK TTP taxonomy mapping, and active defense containment workflows.

B. Critical Societal & Enterprise Needs:

- Enterprise & Healthcare Infrastructure Defense: Small-to-medium enterprise (SME) SOCs, public hospitals, and municipal utilities cannot afford \$100k+/year enterprise SOAR tools (Splunk, Cortex XSOAR), leaving critical infrastructure vulnerable to ransomware.
- Legal Evidence Integrity & Chain-of-Custody: Law enforcement cyber crime cells and air-gapped defense labs require local, tamper-evident JSONL audit logging and courtroom-admissible PDF executive briefs.

C. Bridging the Societal Gap through Project SENTINEL:

- Democratizing Enterprise Security: Providing an open-source, privacy-first AI co-pilot that enables resource-constrained organizations to automate routine threat triage with zero operational cloud licensing overhead.

- Prepare a Passion CV to identify potential entrepreneurial opportunities.

PASSION CV & ENTREPRENEURIAL PROFILE - GOKULA KANNAN M (TEAM LEAD):

- Role & Designation: Systems Architect & Project Lead | Student ID: SEC25CS196 | Department of Computer Science & Engineering (Section A)
- Technical Domain & Core Specialization: Distributed Systems Engineering, AI Pipeline Architecture, Cybersecurity Policy Enforcement, Venture Strategy.
- Key Accomplishments & Leadership Contributions:
 1. System Architecture: Lead architect of Project SENTINEL's 3-Tier AI Router and Zero-Trust Data Sanitizer pipeline.
 2. Venture Pitching & National Validation: Selected to represent Sairam Institutions at Hac'KP 2026 (7th Edition National Hackathon by Kerala Police Cyberdome @ Zoho Corporation Campus).
 3. Research & Academic Output: Co-author of academic literature survey analyzing 6 peer-reviewed IEEE papers on threat triage and privacy-preserving AI.
- Entrepreneurial Opportunity & Commercialization Roadmap:
 - Opportunity Identified: Commercializing SENTINEL as a hybrid open-core / MSSP platform tailored for air-gapped defense labs, municipal SOCs, and police cyber forensic units requiring 100% data sovereignty.